

❖ Week-4 or final report for Botbrain assignment:

1. Test Cases:

3 sample routes:

- 1: Hostel Block-1 → Library
- 2: Main Gate → Auditorium
- 3: Admin Block → Canteen

2. Link of website : <https://preview--hanmanths-campus-compass.lovable.app/>

3. Expected Outcome

The project is expected to deliver a working prototype of a campus navigation chatbot capable of:

- Providing optimal routes between any two locations on the campus.
- Showing estimated distance, walking time, and building timings.
- Answering common FAQs by retrieving information from a database.
- Offering a simple mobile-friendly/chatbot interface for easy use.

4. Applications

- Student & Visitor Navigation: Helps newcomers find locations.
- Quick Information Access: Provides building timings, services and event locations on demand..

5. Conclusion

This project demonstrates how AI search algorithms combined with a database and a chatbot interface can solve real-world navigation problems inside a university campus. The final system not only computes efficient routes but also supplies building information dynamically. Experimental results show that the A* algorithm gives optimal paths with fewer explored nodes, making it the most suitable for such navigation systems.

References

1. Chanakya University map : https://www.google.com/maps/@chanakya_university
2. Loveable website : <https://lovable.dev>